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LPRE.DE announces the release of a new version of [Rocket Propulsion Analysis \(RPA\)](#), a free tool for analysis of liquid propellant rocket engines.

RD-170 (11D521) and RD-171 (11D520) liquid-propellant engines

Abstract

The RD-170/171 (Rocket Engine-170/171) are the world's most powerful liquid propellant rocket engines, designed and produced by NPO Energomash for use with Energia and Zenit launch vehicles. Engines use high performance staged combustion cycle with oxygen rich turbine drive gases. The article provide an information about the development history, key design parameters and operating characteristics of the engines.

Background

By the beginning of work on the 11D520 and 11D521 engines, NPO Energomash (former names OKB-456 and KB EM) had experience in creating high-pressure engines in compressor stations, built in a closed circuit and operating on high-performance components (AT and UDMH).

In particular, 15D119 (RD-263/264) engines with a thrust of $P_z = 1040$ kN (106 tons) and a pressure of 20.6 MPa were created for ballistic missiles. and 15D168 (RD-268) with a thrust of $P_z = 1147$ kN (117 t) and with a pressure in the CS of 22.6 MPa. In the process of working on these engines, the design bureau plant improved the technology of steel casting of complex load-bearing parts (for example, pump casings and automation units, which were previously made of non-ferrous metals). To eliminate the occurrence of combustion instability, plastic antipulsation valves were introduced in the LRE chamber baffles installed on the mixing head and contributed to the attenuation of pressure pulsations.

A certain reserve was also provided by the development of the 8D420 (RD-270) engine with a thrust of 640 tons and a pressure in the CS of 26.1 MPa, operating according to the "gas-gas" scheme. Among other things, special parking seals were developed for this engine to ensure multiple starts, and to reduce the weight and dimensions of the VNA, the design of booster pumps was developed with the location of the turbine blades directly on the impeller-screw of the pump.

The experience of design and experimental testing of large-scale engines and units operating at pressures up to 60 MPa, as well as the mastered technologies for the manufacture of such units, were used when working on the 11D520 and 11D521 engines.

A brief chronology of the creation of the engine

1973 - Start of design and calculation studies and design studies. Technical proposals were developed for the RD-123 engines with a thrust of 800 tons and the RD-150 with a thrust of up to 1500 tons.

1974 - the beginning of experimental research on the development of chemical ignition, mixture formation in the combustion chamber and gas generator, high-frequency stability of the working process in the chamber and gas generator, reusable start-up.

The research was carried out on an analogue engine created on the basis of the serial 15D168 engine. The engine was converted to run on liquid oxygen and kerosene. On new components at a CS pressure of 20 MPa The engine developed a thrust of 90 tons.

In total, more than 300 tests were carried out on 200 copies with a total operating time of 20 thousand s. Based on the results of the research, the following were recommended: a five-fold reserve of resource, a reserve for unloading the axial force at least 20% on the VTA shaft, excluding the possibility of pumps operating in the zone of partial cavitation, and pulsation pressure should not exceed 1-2% of the operating level. Particular attention was paid to the resistance of materials in an oxidizing gas environment.

A control and technological test was introduced for each engine, without overhaul.

February-March 1976 - The government decided to develop the Energia-Buran and Zenit rocket systems.

November 1976 - A preliminary design of the RD-170 engine with the following characteristics was developed: thrust on the ground 740 tons, in the void - 806.4 tons; specific impulse on the ground 309.3 s, in the void - 337 s; the pressure in the combustion chamber is 250 atmospheres, in the gas generator - up to 583 atmospheres; Turbine power up to 297,260 hp

August 25, 1980 - The first firing test of the RD-171 engine (a version of the RD-170 engine for the Zenit launch vehicle) took place.

June 9, 1981 - the first firing test of the RD-171 No A15 engine, which successfully worked out the specified 150 s according to the planned test program.

June 26, 1982 - the first firing test of the RD-171 No A18 engine at the NIIhimash stand as part of the first stage of the Zenit LV. The launch is over an accident that destroyed the only stand in the USSR suitable for testing a rocket stage

with an engine of such power.

May 1983 - the first successful firing test of the RD-171 engine at nominal mode.

December 1, 1984 - the first successful firing test of the RD-171 engine at the NIIhimmash stand as part of the first stage of the Zenit LV.

April 13, 1985 - the first launch of the Zenit LV with the RD-171 engine as part of the first stage, which ended in an accident due to the failure of the fuel consumption control system of the second stage.

June 1985 - the second launch of the Zenit LV with the RD-171 engine as part of the first stage, which ended in an accident as a result of deviations in the operation of the steering engines of the second stage and the subsequent explosion at the end of the active phase.

October 1985 - the third (first successful) launch of the Zenit LV with the RD-171 engine as part of the first stage.

November 1985 - the first bench test of the RD-170 engine as part of Block A of the Energia launch vehicle. The engine successfully worked for the specified time.

May 15, 1987 - First launch of the Energia LV with RD-170 engines as part of the first stage.

December 1987 - Completion of flight tests of the Zenit LV with the RD-171 engine as part of the first stage.

November 15, 1988 - the second launch of the Energia LV with RD-170 engines as part of the first stage.

October 4, 1990 - the fifteenth launch of the Zenit LV, which ended in an accident on the third second of flight as a result of an explosion of the first stage engine. According to the conclusion of the emergency commission, the engine failure occurred due to the destruction of the gas duct swing unit of the second chamber. The most probable cause of the fire was the ingress into the inner cavity of the swing unit of the ignition initiator in the form of a substance of organic origin with concentrated heat release during combustion of more than 30 kilojoules.

July 27, 1991 - The sixteenth launch of the Zenit LV did not take place due to the failure to start the first stage engine.

May 20, 1997 - the twenty-eighth launch of the Zenit LV, which ended in an accident at the 49th second of the flight, as a result of an emergency shutdown of the first stage engine. The rocket fell 28 kilometers from the launch complex.

September 9, 1998 - launch of the Zenit LV, which ended in an accident at the 72nd second of flight.

March 25, 2004 - Delivery of the first commercial RD-171M engine for the Zenit-3SL LV for the Sea Launch program.

July 5, 2004 - The RD-171M engine is certified for use as part of the Zenit-3SL LV for the Sea Launch program.

February 15, 2006 - First launch of the Zenit-3SL LV with RD-171M.

April 25, 2006 - by the decision of the Ministry of Defense of the Russian Federation and Roscosmos, the RD-171M engines were assigned the letter "O1", which allows the use of the engine as part of the Zenit-M launch vehicle in the implementation of state programs.

January 30, 2007 - The launch of the Zenit-3SL LV from the Odyssey offshore launch platform ended in an accident. The rocket exploded at launch. The Russian-Ukrainian interdepartmental commission to investigate the accident found that the cause of the accident was the ignition of a metal particle accidentally introduced from the outside in the oxidizer pump.

June 29, 2007 - the first launch of the Zenit-M LV with the RD-171M engine, carried out from the launcher No 1 pl. No 45 of the Baikonur State Test Cosmodrome.

April 26, 2008 - the first launch of the Zenit-3SLB LV with RD-171M under the Land Launch program, carried out from the Baikonur State Test Cosmodrome.

December 25, 2009 - during the firing acceptance test of the RD-171M engine at the stand of JSC NPO Energomash, the engine failed with the destruction of the engine material part and individual systems of the stand. The commission, having considered the results of processing telemetry measurements and the results of the analysis of the condition of the materiel, concluded the following: the engine failure during its operation at the nominal thrust mode occurred as a result of a fire for 17.6 seconds in the gas cavity behind the TNA turbine at the junction of the turbine exhaust manifold with the gas path of one of the chambers (chamber No2). The development of combustion led to the destruction of the engine gas path behind the turbine for 18.2 seconds, an explosion and a fire. The initiation of the fire occurred as a result of the ingress of a foreign object (substance) into the gas cavity during the manufacture of the engine, which could not be detected by the standard inspection and control procedures. The organic nature of the foreign object (substance) that got inside the gas path in the process of work with the subassembly of the "gas duct block" of the engine was recognized as the most probable.

General information

The engine is made according to a closed circuit with afterburning of oxidizing generator gas after the turbine. Fuel components: oxidizer - liquid oxygen, fuel - kerosene.

The engine consists of four combustion chambers, a turbopump unit (TNA), a booster fuel pump unit (BNAG), oxidizer booster pump unit (BNAO), two gas generators, automation control unit, cylinder unit, automation drive systems (SPA), steering actuator systems (PSA), fuel flow regulator in the gas generator, two oxidizer chokes, fuel choke, oxidizer and fuel start-off valves, four ampoules with starting fuel, starting tank, engine frame, bottom screen, emergency protection system sensors, two heat exchangers for helium heating for pressurization of the oxidizer tank.

One of the main design features of this engine is the presence of four chambers swinging in two planes, and two gas generators operating for one turbine. Four combustion chambers made it possible to have the parameters of the chamber in terms of thrust, close to the mastered range: 185 tons of thrust against 150 tons achieved in other developments. In addition, the presence of four cameras and two GGs makes it possible to organize autonomous testing of these units.

The turbopump unit is located between the chambers, and its axis is parallel to the axis of the chambers. This solution allows you to optimally place the engine within the limited dimensions of the LV tail compartment.

To ensure the maintainability of the structure, detachable flange connections are widely used. Self-sealing flanges are used to ensure the tightness of large-diameter stressed flanges. Double-barrier seals with metal gaskets.

During the development of the engine, it was provided for the possibility of at least twenty times its use as part of the carrier, including inter-flight firing checks as part of the unit. Guaranteed reserves of engine operability in terms of service life and number of switch-ons, in excess of those required for operation (before the last use), must be at least 5 necessary for one flight.

At the end of the 80s, the maximum number on one copy of the engine was 21 tests.



Fig.1. RD-170 engine (without steering gears; the image enlarges when pressed) [1]

Table 1. Technical Parameters of the Engine

Parameter	Significance	Units
Thrust		
near the Earth	740 000	kg
	7256	kN
in the void	806 000	kg
	7904	kN
Thrust throttling limits	100-40	%
Thrust Specific Impulse		
in a vacuum	337	c
at sea level	309	c
Combustion Chamber Pressure	24.5	MPa
Fuel Component Consumption through Engine	2393	kg/s
Component ratio ratio	2.63	m(approx)/m(g)
Adjusting the ratio of components	±7	%
Opening hours	140-150	c
Engine weight		
dry	9755	kg
poured	10750	kg
Dimensions		
Height	4015	mm
Diameter in the nozzle cut plane	3565	mm

Design description

Pneumohydraulic diagram

Description

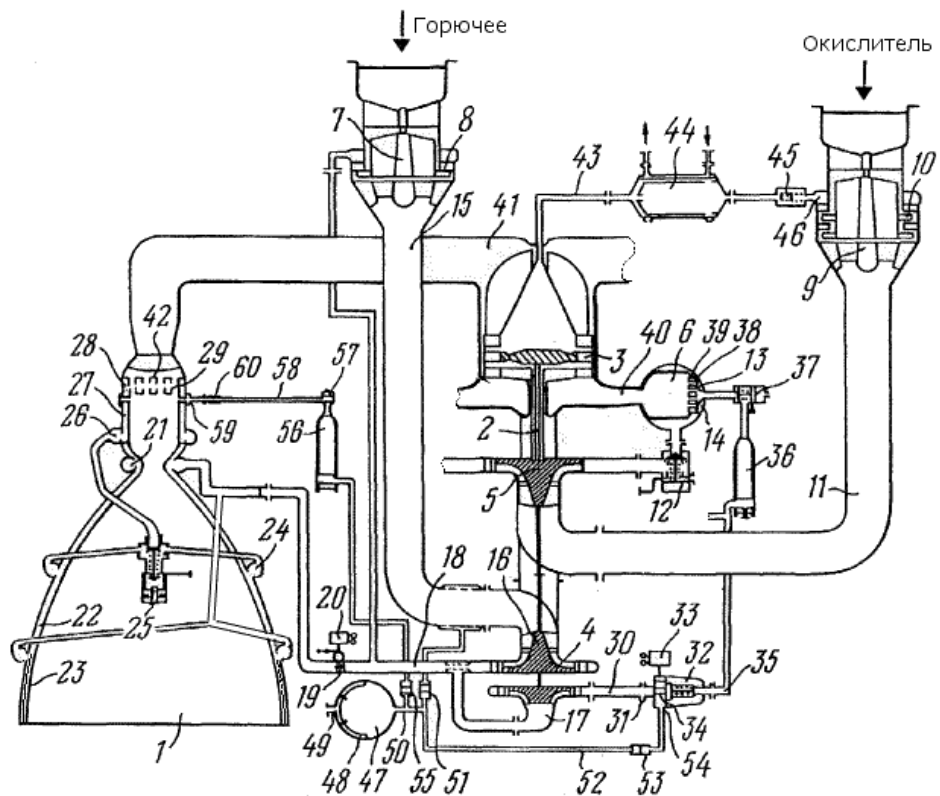


Fig.2.1. Pneumohydraulic diagram [11]
(adapted; conventionally shown 1 CS and 1 GG)

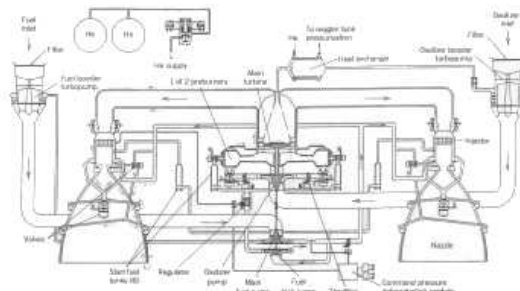


Fig.2.2. Pneumohydraulic diagram [2]
(2 KS are conventionally shown; the image enlarges when pressed)

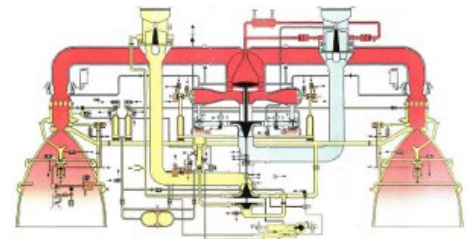


Fig.2.3. Pneumohydraulic diagram
(adapted; conventionally shown 2 KS; the image enlarges when pressed)

The engine comprises a combustion chamber 1, a turbopump unit 2 consisting of a turbine 3, a two-stage fuel pump 4 and a single-stage pump oxidizer 5, two gas generators 6, fuel booster pump 7 driven by a hydraulic turbine 8 and oxidizer booster pump 9, driven by a gas turbine 10.

The oxidizer booster pump (BNAO) 9 is connected through pipeline 11 to the inlet of the oxidizer pump 5, the outlet of which is connected to the manifold valve through the shut-off valve 12 cavity 13 of the mixing head 14 of the gas generator 6. An oxidizer filter is installed at the inlet of the BNAO.

The fuel booster pump (BNAG) 7 is connected through the pipeline 15 to the inlet of the first stage 16 of the fuel pump 4. The first stage of the fuel pump 16 is connected to the inlet of the second stage 17 of the fuel pump and through the pipeline 18, in which the choke 19 with an electric drive 20 is installed, connected to the commutator 21 of the combustion chamber 1, from which the fuel is distributed through the regenerative cooling channels 22 of the combustion chamber 1. A fuel filter is installed at the BNAG inlet.

The regenerative cooling channels 22 of the nozzle 23 through the manifold 24 are connected to the start-off valve 25. The outlet of this valve is connected to the manifold 26, located on the cylindrical part of the combustion chamber. Outlet of the manifold 26 through the regenerative cooling channels 27 of the cylindrical part of the combustion chamber is connected to the fuel cavity 28 of the mixing head 29 of the combustion chamber 1.

The second stage 17 of the fuel pump 4 (through which 20% of the total fuel flow passes) is connected to the main inlet through the pipeline 30 31 of the thrust regulator 32, controlled by an electric actuator 33 and having a check valve 34 at the inlet. The output 35 of the thrust regulator 32 is connected to ampoules 36 (2 pcs.), filled with starting fuel triethylaluminum $\text{Al}(\text{C}_2\text{H}_5)_3$. Outlets from these ampoules through start-off valves 37 38 mixing heads, 39 gas generators, 6 are connected to the fuel cavity. The outlet of the gas generators 40 is connected to the turbine 3, the outlet of which through the Piping 41 is connected to the cavity 42 of the mixing heads 29 of the combustion chambers 1.

In addition, the outlet of the turbine 3 through the piping 43, in which the heat exchanger 44 and the pressure valve 45 are installed, is connected to the turbine manifold 46 drives of the booster pump 9 oxidizer.

The pneumatic-hydraulic scheme of the LRE also contains a starting system, which includes a starting tank 47 with a diaphragm seal 48, a gas supply pipe 49 high-pressure pipe and outlet 50. The outlet pipe 50 of the starting tank 47 is connected to the pipeline 15 through the filling valve 51 fuel supply from the fuel booster pump 7. In addition, the outlet 50 on one side through the pipeline 52, in which the The check valve 53 is connected to the second inlet 54 of the thrust

regulator 32, through which the engine is started, and on the other side through the return valve Valve 55 is connected to an ampoule 56 filled with starting fuel (hypergolem), the outlet of which through valve 57 is connected to the line 58 of the launch fuel supply fuel to the ignition injectors 59 of the combustion chamber. A nozzle 60 is installed in the line 58, which provides a metered supply of starting fuel to the ignition injectors.

To reduce the after-action impulse, fuel shut-off valves are installed between the cooling ducts of the nozzle and the combustion chamber (valves 25), as well as in front of the collector of the second and third chords (shown in Fig. 2.2).

The pneumatic valves are actuated by helium from the high-pressure cylinder block by means of solenoid valves.

Engine Operation

The engine is started according to the "self-start" scheme. The actuators 20 and 33 are pre-positioned for initial installation thrust regulator 32 and throttle 19. Then the tank valves of the rocket are opened (not shown in the diagram) and under the influence of hydrostatic head and pressure pressurization fuel components fill the cavities of the oxidizer and fuel pumps up to the start-off valves 12 and 25 and the check valve 34 of the thrust regulator 32 respectively. The engine cavities are filled with fuel up to the starting ampoules 36 and 56 through the filling valve 51, check valves 53 and 55. The starting tank 47 is also filled with the main fuel. This state is considered to be the starting point for starting the engine.

When the engine is started, the reservoir 47 is pressurized and the fuel is expelled from it, the pressure of which breaks through the membranes (not shown) of the starting ampoules 36 and 56. At the same time, start-off valves 12 and 37 and 25 shall be opened, respectively. As a result, the starting fuel from ampoules 36 and 56 under the influence of pressure created by the starting tank enters the gas generators (through open valves 37) and chambers (through check valves 57). The starting fuel entering the gas generators is ignited with oxygen, which also enters the gas generators due to the pre-launch pressurization of the rocket tanks and hydrostatic head in them. Fuel, having passed through the cooled path of the combustion chambers, enters the mixing heads of the chambers after a fixed time Combustion 1. During this time of delay, the combustion process in the gas generators manages to begin and the generated generator gas spins up the turbine 3 TNA 2. After the turbine, the oxidizing gas enters through four cooled gas ducts 41 into the mixing heads 29 of the four combustion chambers, where it ignites with starting fuel coming from the ignition injectors 59 and subsequently burned with the fuel entering the chambers. Arrival time of both components in the combustion chambers is selected in such a way that TNA 2 has time to enter the operating mode, while back pressure has not yet been established in chambers 1.

As the pressure behind the fuel pump 17 increases, the starting tank 47 is automatically switched off by closing the check valves 53 and 55. And the fuel supply of gas generators 6 is switched to pump 17 due to the software opening of the throttle of the thrust regulator 32.

Part of the oxidizing gas from the turbine outlet is taken to drive the two-stage gas turbine 10 of the booster prepump 9. This gas, passing through heat exchanger 44, heats the gas used to pressurize the rocket tanks. After turbine 10, the gas is discharged into the outlet manifold 11, where it is mixed with main oxidizer flow and condenses. Use of gas taken from the outlet of the TNA turbine as the working fluid of the turbine drive booster pump of the oxidizer allows you to reduce the temperature in the gas generator and, accordingly, reduce the power of the TNA turbine.

Part of the fuel from the outlet of the pump 4 goes to the drive of the single-stage hydraulic turbine 8 of the fuel booster pump 7.

A small part of liquid oxygen is taken from the manifolds of gas generators and enters the cooling path of the casing turbines and gas pipelines.

At the entire stage of starting the engine, the opening of the throttle of the thrust regulator 32 and the fuel throttle 19 from the positions of the initial Positions corresponding to the rated mode of the engine by means of appropriate drives 33 and 20.

Thus, a smooth start of the engine is carried out with access to the main mode in 3 seconds.

Before shutdown, the engines are switched to the final stage mode, which is 50% of the nominal one.

Table 1a. Simplified cyclogram of operation of the 11D521 engine as part of Block A of the Energia launch vehicle (according to the flight program on November 15, 1988)

Nº	Time (s) from the start command ("ascent contact")	Description (Condition)
1	-3.2	Launch, a software set of starting thrust.
2	-0.2	Reaching the main stage of thrust.
3	38	Start of software throttling to reduce velocity head.
4	74	End of program throttling to reduce velocity head.
5	108.5	Start of software throttling to limit the longitudinal overload to 2.95 units.
6	130	Transfer of the engine to the final thrust mode of 49.5%.
7	142	Engine shutdown.

Camera

The chamber is a soldered-welded one-piece unit and consists of a mixing head, a combustion chamber and a nozzle. The chamber is attached to the gas path using a flange connection.

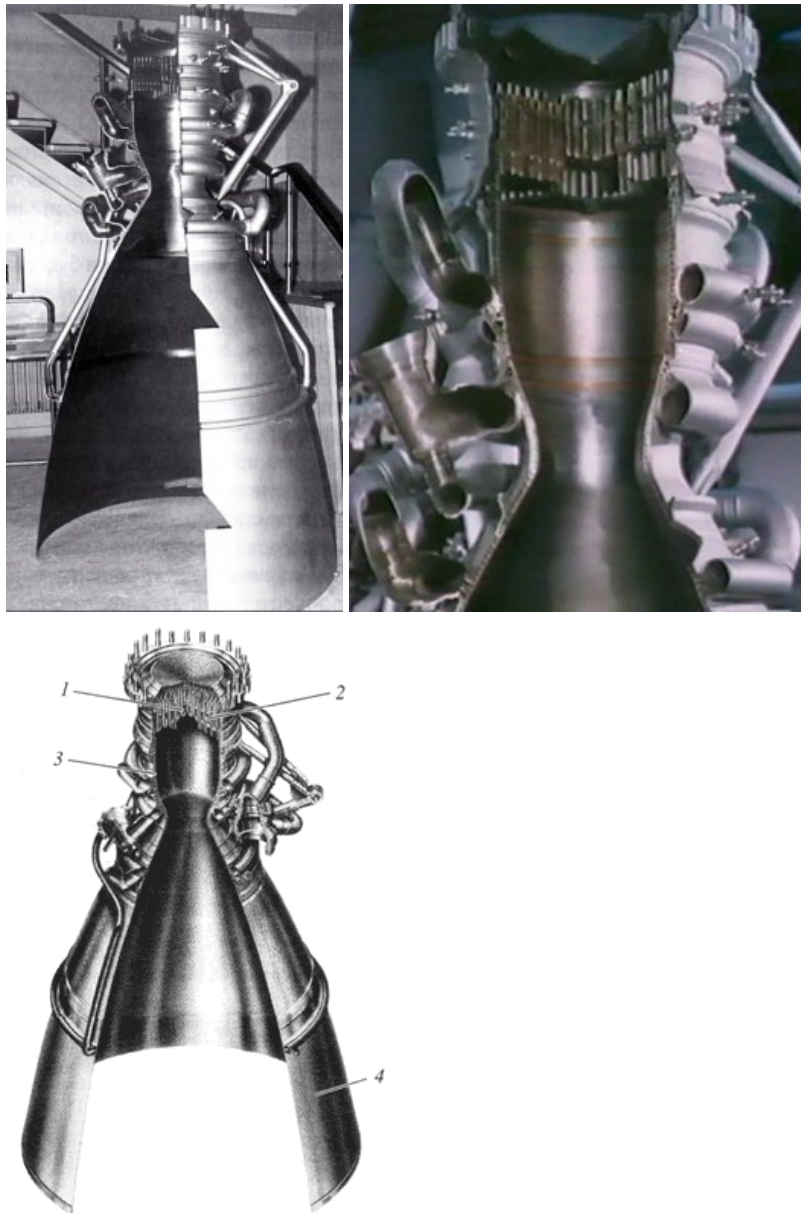


Fig.3. Camera [18],[15],[3]
(image on the left enlarges)

1. Mixing Head
2. Nozzle
3. subcritical part of the combustion chamber
4. Supercritical part of the combustion chamber

Table 2. Technical parameters of the camera

Parameter	Significance	Units
Reduced Length of CS	1079.6	mm
Diameter of the CS	380	mm
Diameter of the minimum nozzle section	235.5	mm
Degree of narrowing of the subsonic part of the nozzle	2.6	
Nozzle outlet diameter	1430	mm
Degree of expansion of the supersonic part of the nozzle	36.87	
Chamber length	2261	mm
Temperature in CS	3676	K
Pressure in the CS	24.5	MPa
Nozzle outlet pressure	0.072	MPa
Thrust coefficient		
in a vacuum	1.86	
at sea level	1.71	
Camera deflection angle	8	Degrees

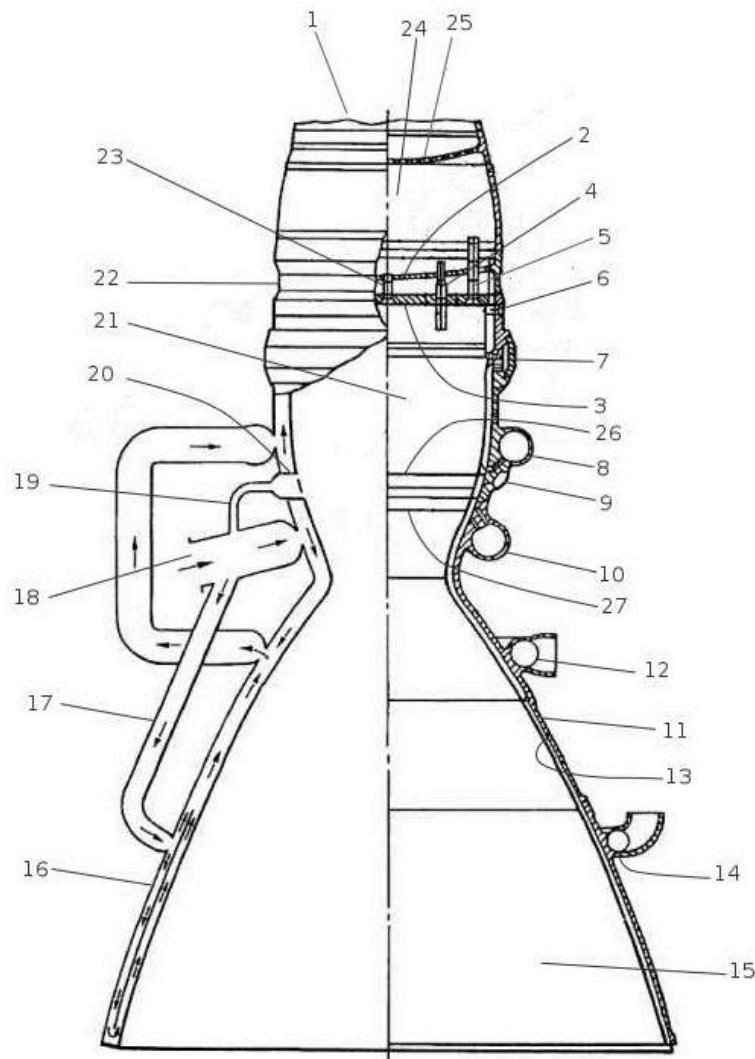


Fig.4. Diagram of fuel supply to the cooling duct of the chamber [8]:

1. Gas pipeline
2. Middle bottom of the mixing head
3. Front (firing) bottom of the mixing head
4. Anti-pulsation baffle nozzles (54 in total)
5. Main Injectors
6. ignition mixture supply (4 nozzles fed from a separate manifold)
7. Upper Curtain Chord Manifold
8. fuel supply manifold for cooling the cylindrical part of the compressor station
9. Collector of the middle 26 and lower 27 curtain belts
10. main manifold of fuel supply to the compressor station
11. CS External Power Wall
12. Fuel drainage manifold from nozzle cooling path
13. CS Inner Wall
14. fuel supply manifold for cooling the nozzle outlet
15. Nozzle
16. Fuel moves to the nozzle cut through even (conditionally) channels and returns through odd channels
17. fuel supply to cool the nozzle outlet
18. fuel supply from the pump
19. fuel supply to the middle and lower belts of the curtain
20. Partition in canals
21. Cylindrical CS Part
22. Mixing Head
23. Central nozzle
24. Gas cavity of the mixing head
25. Gas pipeline

- 25. Perforated rear bottom of the mixing head
- 26. Middle Veil Belt
- 27. Lower Veil Belt

The chamber body consists of a combustion chamber and a nozzle. The chamber body includes an outer power shell 11 and an internal firing wall 13 with milled channels forming the external regenerative cooling path of the chamber, which has three cooler inlets. The first inlet is connected to the cooling path of the critical section of the nozzle, the second inlet is connected to the cooling path of the outlet part of the nozzle, and the third - with the cooling path of the combustion chamber. At the same time, the first exit is connected to the third entrance, and the first entrance, the second entrance and the supply to the two lower ones belts of slotted curtains are united by a common branch pipe, branched and placed outside the chamber.

Internal cooling is provided by three belts of slot curtains in the subcritical part of the combustion chamber. Through them, the wall is supplied about 2% of the fuel in the form of films that evaporate and protect it from heat flows, which in the critical section of the nozzle reach values of the order of 50 MW/m^2 .

The ignition means are made of four equally circumferentially spaced jet nozzles 6 installed behind the front (firing) bottom 3 in the power camera body 11. The axes of the flow ports of the jet nozzles are located at an acute angle to the exit from the power body and are deflected in a circle in the transverse plane from the longitudinal axis of the load-bearing body in the same direction, and the axis of the flow port of each jet nozzle is crossed in relation to the axes of the flow holes of the injectors adjacent to it. The injectors are hydraulically connected by a common manifold.

All injectors are two-component with axial oxidation gas supply and tangential fuel supply. Nozzles located near of the fire (inner) wall of the chamber are made with increased water resistance along the fuel line in comparison with others injectors by reducing the diameters of the fuel supply holes, i.e. providing reduced fuel consumption in comparison with other injectors.

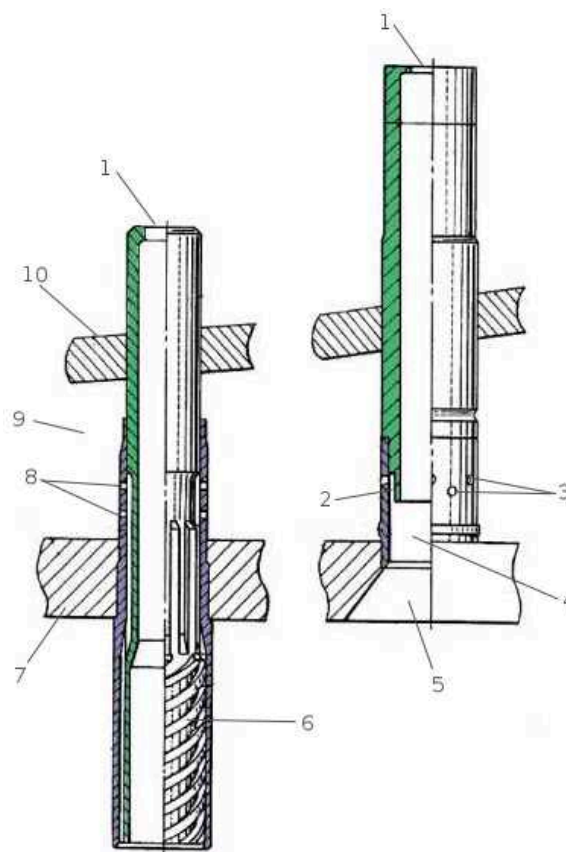


Fig. 5. Nozzle diagram [8]:

- 1. Generation gas supply
- 2. a cavity between two walls into which fuel is supplied
- 3. tangentially located holes for fuel supply
- 4. Primary mixing area
- 5. Outlet cone-shaped part of the nozzle
- 6. Screw swirler
- 7. Front (firing) bottom of the mixing head
- 8. Fuel Supply Holes
- 9. fuel cavity between the front and middle heads of the mixing head
- 10. Middle bottom of the mixing head

To suppress pressure pulsations, the initial zone of mixture formation and combustion, in which, as a rule, high-frequency oscillations, is divided into seven approximately equal volumes by means of antipulsation partitions consisting of protruding the firing bottom of the injectors, which do not fit tightly to each other in their cylindrical constituents. Thanks to this, the natural frequencies of oscillations in the volumes between the baffles, shifting far from the resonant frequencies of the combustion chamber structure. In addition, protruding nozzles stretch the combustion zone, which also reduces the possibility of high-frequency phenomena. Gaps between loosely fitting protruding nozzles have an additional damping effect.

The part of the injector protruding beyond the firing bottom is cooled by fuel passing through the spiral channels (screw swirler) 6 of the inner bushing.

The rest of the nozzles are buried in the firing bottom (their outlet cavities 4 go into conical bores 5 in the firing bottom 7) and are made with various hydraulic resistance during fuel supply with division by mass fuel consumption into three groups with the possibility of providing the difference in fuel consumption between each group is from 3% to 10% in the nominal mode. At the same time, the injectors (except for those located near the firing chamber walls) are fixed in the firing bottom and middle bottom so that nozzles from different groups are adjacent to each other by cyclic sequential spiral repetition of the nozzle arrangement from the first to the last group.

The introduction of injectors with different flow rates is necessary in order to reduce the effects of high-frequency oscillations in the operating modes of the engine.

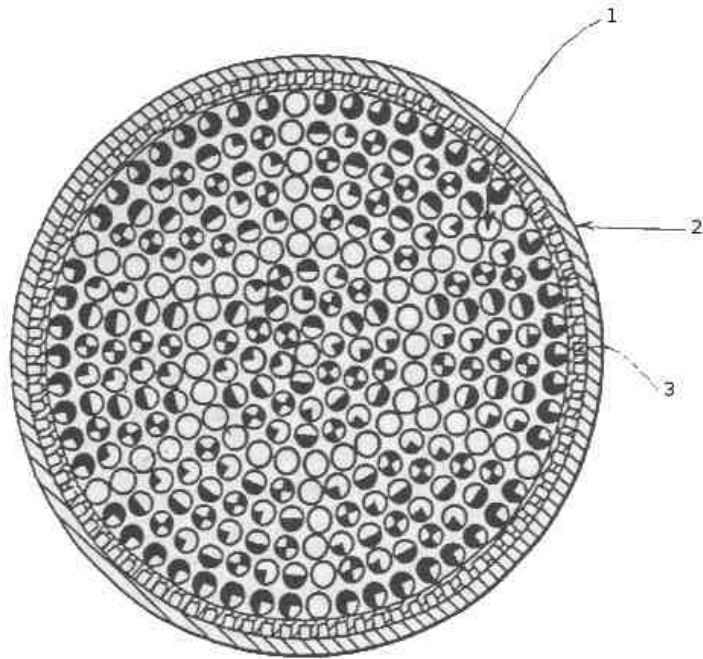


Fig.6.1 Nozzle layout on the mixing head [8]:

1. Nozzles that form anti-pulsation partitions
2. external power wall of the CS, soldered to the fire wall
3. Fire wall with milled channels

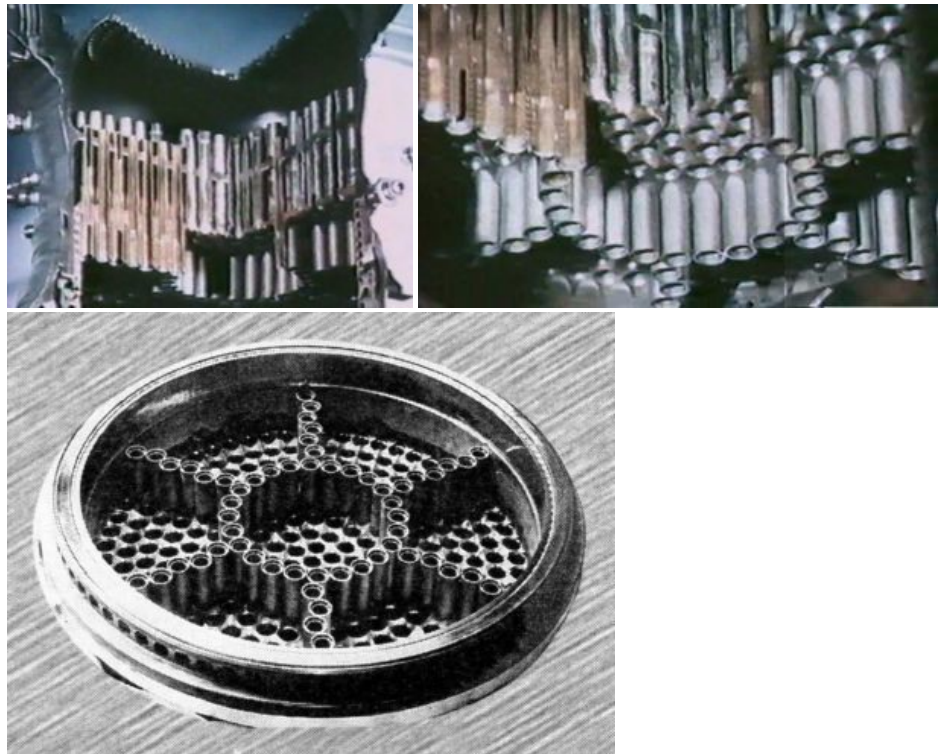


Fig.6.2 Location of nozzles on the mixing head (images are enlarged) [15], [19]

Swing Node

Each of the four chambers is equipped with a swing unit. The thrust force is transmitted from the camera to the power frame through the gimbal. The generator gas fired on the turbine is supplied to the compressor station through a 12-layer composite bellows placed in the inside the gimbal. The bellows is armored with special rings and cooled with a small amount of cold oxygen, flowing between the inner surface of the bellows and the thin inner wall.

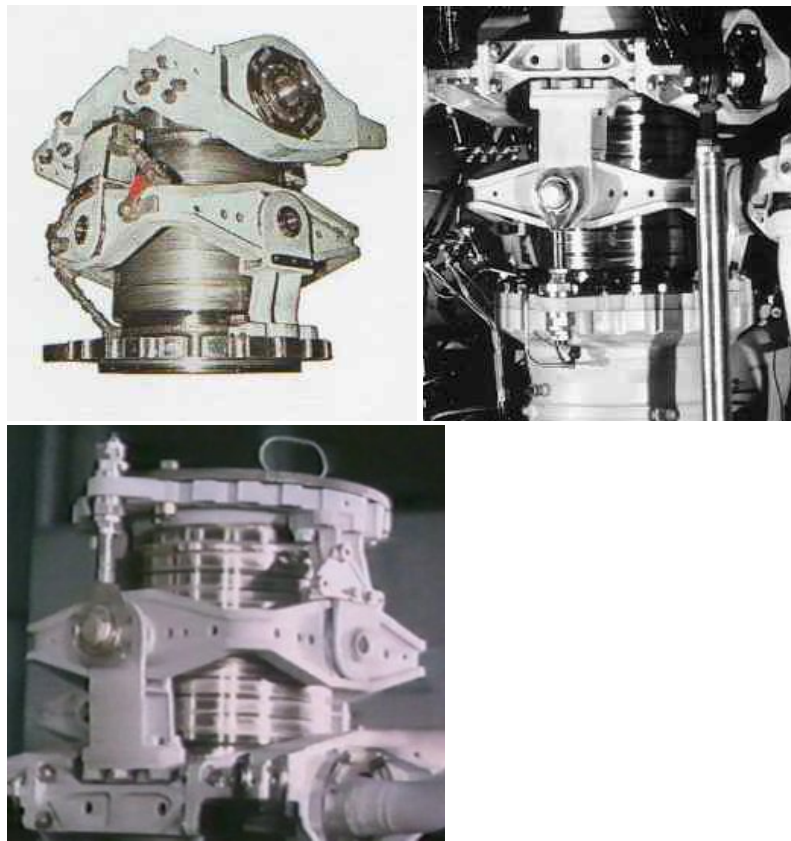


Fig.7. Appearance of the swing unit [5],[1],[15]

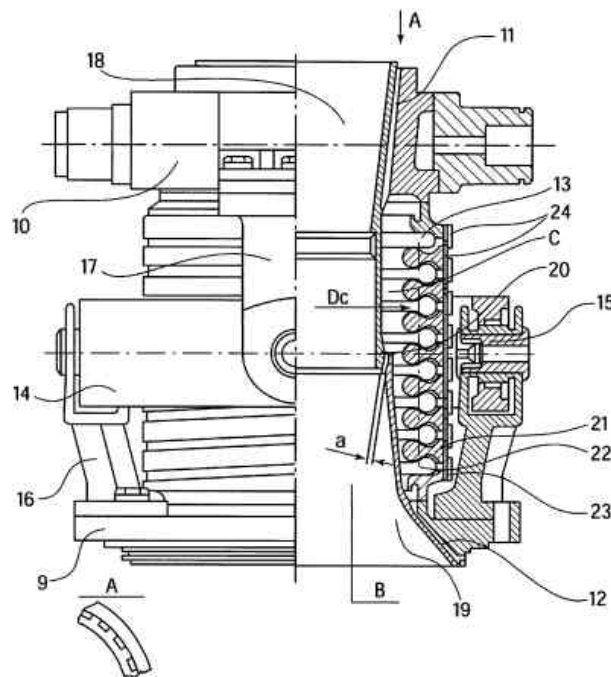


Fig.8. Diagram of the swing unit [10]

The oscillation unit consists of support rings 9 and 10, which are respectively hermetically connected to the combustion chamber and the gas duct (turbine outlet), which contain external flow cooling consumables 11 and 12, also shown in Form A. The bellows 13 is located inside the gimbal ring 14. Cardan ring 14 through hinges 15 forming two rotary axes, connected by power brackets 16 and 17 with support rings 9 and 10.

Inside the bellows 13 there are two shells 18 and 19, each of which is a body of rotation and is cantilever attached respectively to one of the said support rings, whereas, the free end of the sheath 18 is made in the form of a nipple with a spherical end 20 and is installed with a gap a in the sheath 19. The center of the nipple sphere with the spherical end 20 is located on the axis of swing of the chamber. The value of the specified clearance is chosen to ensure the flow rate cooling working fluid (oxidizer) necessary for reliable cooling of the bellows 13.

The bellows 13 is multi-layered and is equipped with protective rings 21 inserted between the corrugations 22 of the bellows 13. Outside the protective rings 21, a tight-fitting casing 23 is installed, made of layers of cylindrical spirals 24, connected by the ends to the support rings 9 and 10 of the bellows assembly. Adjacent layers of spirals are adjacent to each other, and their turns are wound in opposite directions.

The installation of a metal power casing in the form of a metal cylindrical spiral outside the protective rings 21 of the bellows 13 increases the its strength properties and at the same time limits the spontaneous bending of the bellows 13 when the engine chamber is rotated at relatively large angles (10-12°), thereby increasing its stability.

Turbopump unit (TNA)

The turbopump unit is made according to a single-shaft scheme and consists of an axial single-stage reaction turbine, single-stage oxidizer screw pump and two-stage centrifugal fuel pump (the second stage is used to supply part of the fuel to gas generators).

Table 3. TNA

Parameter	Significance Units	
	Oxidizer	Fuel
Pump outlet pressure	60.2	50.6 MPa
Component flow rate through pump	1792	732 kg/s
Impeller diameter	409	405 mm
Pump efficiency	0.74	0.74
Shaft Power	175	77 hp
	600	760
	129.2	57.2 MW
Shaft Speed	13 850	min-1
Turbine power	257 360	hp
	189.3	MW
Turbine inlet pressure	50.9	MPa
Number of turbine stages	1	
Degree of Turbine Pressure Reduction	1.94	
Turbine inlet temperature	772	K
Turbine efficiency	0.79	



Fig.9. TNA [18]
(image enlarged)

On the main shaft with a turbine there is an oxidizer pump, coaxially with which two stages of the fuel pump are located on the other shaft. The shafts of the oxidizer and fuel pumps are connected by a gear spring to unload the shaft from temperature deformations arising as a result of large temperature difference of the pump bodies, as well as to prevent freezing of fuel.



Fig.10. Shaft with turbine, oxidizer pump screw centrifugal wheel, bearings and impeller seals [5]

Effective auto-unloading devices have been developed to protect angular contact bearings of shafts from excessive loads.

Turbine

In a closed-loop oxidation engine, the protection of the TNA oxygen duct units from ignition under the influence of accidental ignition initiators. Due to the exceptionally high pressure in the tract 11D520 and 11D521 engines, as well as high mechanical loads characteristic of a powerful engine, The problem of protection against burning during their creation was especially acute.

To prevent fire due to breakage of structural elements or friction of rotating parts against stationary parts (due to sampling of gaps from deformations or hardening on mating surfaces from vibration) the gap between the blades of the nozzle apparatus and rotor are made relatively large, and the edges of the blades are relatively thick.

To prevent fire and destruction of parts of the turbine gas path, nickel alloys, including heat-resistant ones, are used in the design for hot gas pipelines. The stator and exhaust tract of the turbine are forcibly cooled with cold oxygen. In small places radial or end clearances, various types of heat-shielding coatings are used (nickel for rotor and stator blades, metal-ceramic for the rotor), as well as silver or bronze elements that prevent fire even in case of possible contact rotating and stationary parts of the turbopump unit.

To reduce the size and mass of foreign particles that can lead to a fire in the turbine gas path, at the engine inlet A filter with a mesh of 0.16x0.16 mm was supplied.

Oxidizer Pump

The high pressure of liquid oxygen and, as a result, the increased orpsity of ignition caused the structural Features of the oxidizer pump.

So, instead of floating O-rings on the impeller piles (usually used on less powerful VTAs), fixed slotted seals with a silver lining, since the process of "floating" of the rings is accompanied by friction at the points of contact between the impeller and the housing and may cause the pump to catch fire.

The auger, impeller and torus elbow require particularly careful profiling, and the rotor as a whole requires special measures to ensure that the rotor is particularly profiling. ensuring a dynamic balance in the work process. In the opposite case, due to large pulsations and vibrations Pipelines are destroyed, fires occur in joints due to mutual movement of parts, friction and hardening.

To prevent fire due to breakage of structural elements (auger, impeller and guide vane blades) Under conditions of dynamic loading with subsequent ignition due to rubbing of debris, the following means were used: improvement of structural perfection and strength through geometry, materials and purity of work, as well as the introduction of new technologies: isostatic pressing of cast billets, the use of pellet technology and other types.

Oxidizer Booster Pump Unit (BNAO)

The oxidizer booster pump consists of a high-pressure screw and a two-stage gas turbine, which is driven by oxidizing gas taken after the main turbine and then bypassed to the main pump inlet.

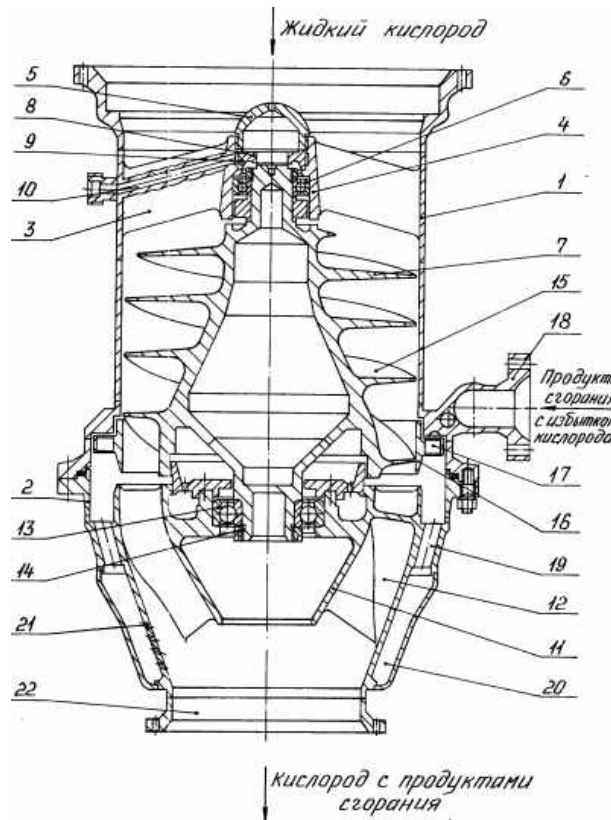


Fig.11a. Simplified diagram of the oxidizer booster pump unit [16] (the image enlarges).

The composite body, consisting of bodies 1 and 2 connected by a flange connection, has a sleeve 4 fixed on the power ribs 3, the inner cavity of which is closed by a fairing 5. Inside the sleeve 4 there is a ball bearing 6, seated on the impeller of the pump, made in the form of a screw 7. The fairing 5 presses the liner 8 installed in the sleeve 4. In the liner 8 there are holes 9 connecting the cavity of the liner 8 with the high-pressure channel 10.

Hull 2 comprises a fairing 11 fixed in it by means of straightening blades 12. This fairing is equipped with a ball bearing 13, fixed with a nut 14 on the auger 7. The auger has 15 blades. Along these blades, the screw is inserted into the impeller of the turbine 16 (which actually consists of two stages, and not one, as shown in the simplified diagram) and welded with it, i.e. the turbine impeller is fixed to the peripheral part of the pump impeller.

The turbine impeller has profiled blades 17, the interblade spaces of which are connected by nozzles in the nozzle apparatus with the inlet manifold. Combustion products with excess oxygen are supplied through the inlet pipe 18. The outlet cavity of the turbine, made in the housing 2 in the form of an annular cylindrical cavity, is connected by channels 19 with a conical annular pipe 20, which is connected by holes 21 to the cylindrical outlet 22.

During the operation of the BNAO, liquid oxygen is supplied to the pump inlet (shown by the arrow), and combustion products with excess oxygen taken from the gas duct after the main VTA turbine (see the PGS in Fig. 2) are supplied to the turbine inlet (shown by the arrow). The combustion products then fall onto the profiled blades 17 of the turbine, providing the supply of liquid oxygen by the screw 7. Behind the turbine, the combustion products enter through the holes 19 into the cavity of the pipe 20, and then through the holes 21 to the pump outlet, where they are mixed with liquid oxygen and condensed. To solve the problem of low-frequency pulsations during gas condensation, crushing of the stream discharging gas is used.

Unloading of the screw 7 from the action of axial forces is provided by the supply of high-pressure liquid oxygen (see Fig. 2.2) through the high-pressure channel 10 into the high-pressure cavity of the auto-unloading device. In the place of a small gap between the impeller and the body in the high-pressure cavity of the automatic unloader, a silver lining is used to prevent fire in case of possible contact.

The first developed "hot gas" valve (45 in Fig. 2.1) operating in the conditions of oxygen generator gas with high temperature and high pressure is installed in the combustion gas supply line to the BNAO turbine.



Fig.11b. Rotor of the booster pump unit of the oxidizer [5]

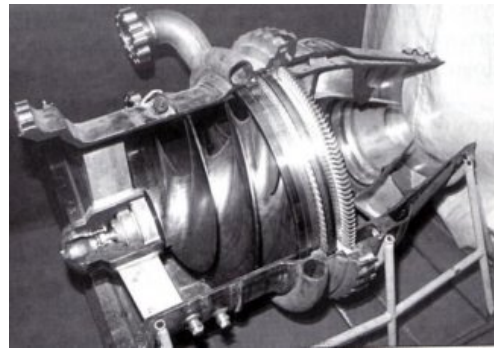


Fig.11c. Booster pump unit of the oxidizer [18] (image enlarges)

Fuel Booster Unit (BNAG)

The fuel booster pump consists of a high-pressure screw and a single-stage hydraulic turbine running on kerosene, taken after the main pump.

Structurally, the fuel booster pump is similar to the oxidizer booster pump with the following differences:

- a single-stage hydro turbine operates on fuel taken from the outlet of the main VTA fuel pump;
- high-pressure fuel is removed to unload the screw from axial actions from the inlet manifold of the BNAG hydroturbine.



Fig.12. Fuel booster unit [5]

Gas generator and gas path (gas ducts)

Table 4. Gas generator

Parameter	Significance	Units
Inner diameter GG	300	mm
Pressure in the MC	53.5	MPa
Fuel consumption through GG	836	kg/s
Component Ratio	54.3	m(approx)/m(g)
Number of MC	2	



Fig.13. Gas generator [9]

Single-zone gas generator producing gas with excess oxidizer for turbine drive, consists of a brazed-welded body with a spherical outer shell and an outlet pipe rigidly connected to it, a cylindrical fire chamber with a diameter of 300 mm and a mixing head equipped with two-component and two-stage for the oxidizer by injectors, the design of which is made with a combustion zone and a gas ballasting zone inside the injectors. In fact, each nozzle forms, together with the channel of a thick-walled firing bottom, in which it is located, Individual dual-zone gas generator. As a result, the uniformity of the temperature field along the transverse the cross-section of the total gas flow formed by such nozzles at high flow intensity.

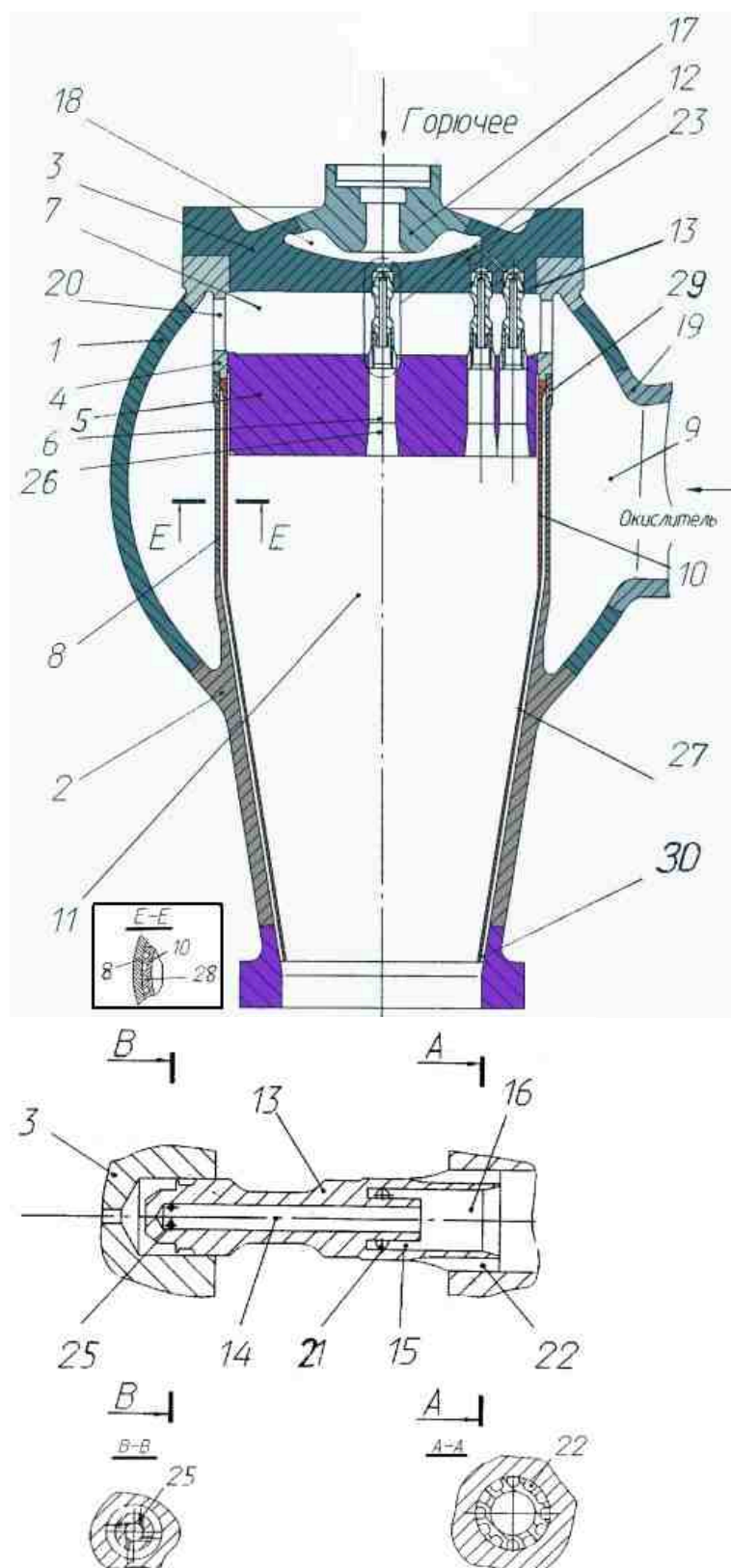


Fig.14a. Diagram of the gas generator [6],[7]:

1 - spherical power shell; 2 - outlet pipe; 3 - cover; 4 - bushing; 5 - firing bottom; 6 - through chambers in the firing bottom; 7 - oxidizer cavity; 8 - spacer (outer wall of the fire chamber); 9 - annular cavity; 10 - shell (inner wall) of the firing chamber; 11 - firing chamber; 12 - mixing module (nozzle); 13 - body of the mixing module; 14 - fuel channel; 15 - annular channel oxidizer; 16 - mixing chamber; 17 - fuel supply pipe; 18 - fuel cavity; 19 - oxidizer supply pipe; 20 - windows in bushing 4; 21 - tangential holes of the oxidizer supply; 22 - grooves on the outer surface of the nozzle body; 23 - calibrated fuel supply channels; 25 - tangential holes of fuel supply; 26 - conical bores; 27 - cooling cavity; 28 - channels forming a cooling cavity; 29 - oxidizer inlet holes into the cooling cavity; 30 - annular slit of the oxidizer outlet from the cooling cavity.

During the operation of the gas generator, fuel from the nozzle 17 fills the cavity 18 and is supplied through calibrated channels 23 and tangential openings 25 into channels 14 and further into mixing chambers 16. The oxidizer is inserted through the nozzle 19 into the annular cavity 9, through windows 20 fills the cavity 7. Part of the oxidizer enters the mixing chamber 16 through tangential holes 21, where, mixing with fuel, it causes its ignition. Through the grooves 22, the oxidizer is also fed into the chamber 6, providing mixing of high-temperature combustion products. Further, in the fire chamber 11, high-temperature products are cooled with simultaneous evaporation of the liquid and heating of the gaseous oxidizer. At the outlet of the gas generator to the products gas generation is mixed with an oxidizer supplied through the annular slit 30.



Fig.14b. TNA with gas generators [19]

The gas generator provides oxidizing gas at the outlet in a wide range of temperatures (from 190 to 600°C), which allows you to adjust the engine thrust from 30 to 105% of the nominal value.

The housing and mixing head are connected by means of a split flange. To ensure tightness, seal with metal gaskets.

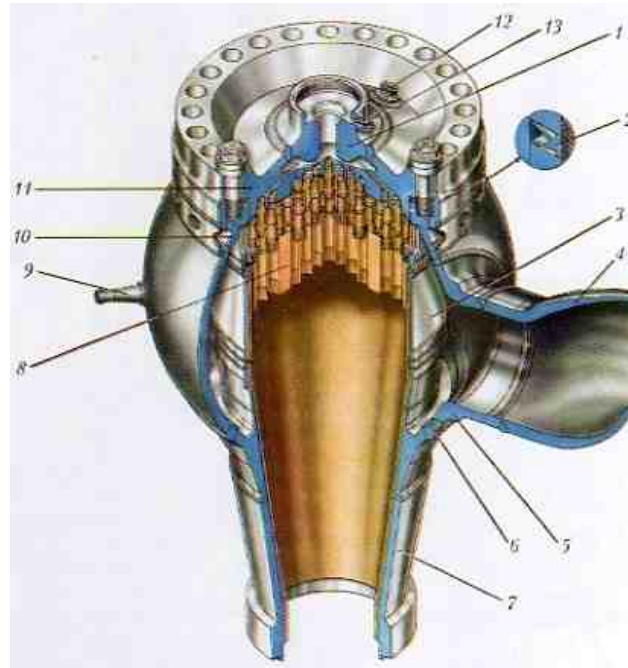


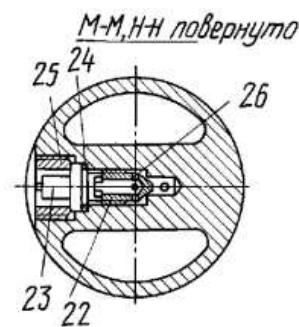
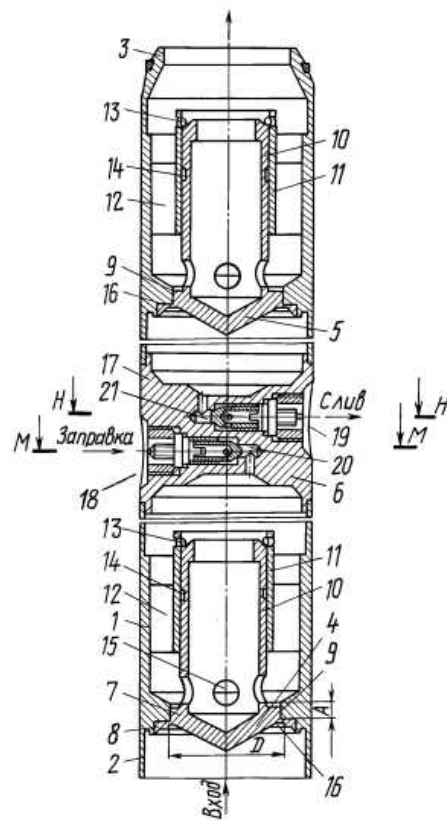
Fig.15. Gas generator [9]:

1. Flange
2. Gasket
3. spacer
4. Pipeline
5. Spigot
6. spherical bottom
7. Closing section
8. The bottom is firing
9. Fitting
10. Ring
11. Body
12. Fitting
13. Fitting

To ensure an acceptable level of temperature stresses in the load-bearing body parts, gas ducts Between the gas generators, turbine and chambers, they are cooled with oxygen.

To prevent fire, increased (in comparison with less powerful engines) the requirements for the cleanliness of gas paths and the prevention of the presence of organic substances.

Ampoule with starting fuel



The ampoule comprises a body 1 with inlet 2 and outlet 3 pipes of membrane units 4 and 5 installed inside the body 1, and a means for filling the body with starting fuel 6. Each membrane assembly 4, 5 comprises a piston 7 which can be configured as a single unit with the membrane 8 or in which the membrane 8 is hermetically connected to its outer surface. The piston 7 is installed in the guide 9 of the body on a sliding fit.

The peripheral section of the diaphragm 8 is hermetically welded with the body 1 under the guide 9. The piston 7 is connected to the shank 10, which may be cylindrical or of some other shape and placed in the sleeve 11 on brackets 12 is attached to the body 1 of the ampoule. The sleeve 11 has a spring retainer 13, e.g., made in the form of a spring ring, and the shank 10 is made with an annular groove 14.

When the diaphragm assembly is actuated, the spring retainer 13 restricts the movement of the shank 10. The shank 10 is made with holes 15 for venting gas from the stagnant zone when filling the ampoule. The diaphragm 8 on the side of the inlet 2 is made thin in the form of an annular bridge 16, which is torn when interacting with the working medium at diameter D. Size D is slightly smaller than the diameter of the piston 7. At the junction of the diaphragm 8 with the piston 7, it is made with a lower thickness in order to exclude seizures when the

Fig.16. Diagram of the ampoule with starting fuel [17] (the image is enlarged).

piston 7 moves in the guide 9 of the body 1.

The design includes a means for filling the body with starting fuel 6, which is installed in the partition 17 of the building 1 and consists of two plugs - the filling plug 18 and the drain plug 19, which are installed in the filling 20 and drain 21 channels, respectively. Each of the plugs has a threaded plug 22, a sealing plug 23, a sealing gasket 24, and a nut 25. The threaded plug 22 has a flow hole 26.

Filling the ampoule with starting fuel is carried out as follows. On the assembled ampoule, before the installation of nuts 25 and sealing plugs 23, the threaded plugs 22 are not completely screwed in, so as to ensure the opening of the passage section of the filling channel 20 and the drain channel 21 through the hole 26. Starting fuel is filled by supplying it through the filling channel 20 to the inner cavity of the body 1 between membrane units 4 and 5, and then through the drain channel to the drain. After the filling of the ampoules is completed, the threaded plugs 22 are screwed in to the stop, after which the starting fuel is drained before the threaded plug 22 of the filling plug 18 and after the threaded plug 22 of the drain plug 19. After that, sealing plugs 23, sealing gaskets 24 and nuts 25 are installed. After that, the ampoule is ready to be installed on the rocket engine. In the inner cavity of the ampoule in the body 1, a gas cushion is formed between the membranes 8 as a result of assembling and filling the ampoule. The presence of a gas cushion helps to ensure the reliability of the ampoule during storage and effective movement with the acceleration of the piston 8 when applying medium pressure to the inlet of the ampoule.

The device works as follows. When the high-pressure component is exposed to the inlet side of the membrane assembly 4, the membrane 8 is deformed and then destroyed along the circumference D. In case of uneven destruction of the membrane 8, with the appearance of leakage, the pressure in front of the piston 7 does not drop, thanks to the work of the throttling slot formed by the guide of the body 9 and the piston 7, the piston 7 continues to move, and after the complete destruction of the membrane 8, it accelerates. The acceleration motion of the piston 7 is due to the presence of a differential pressure force acting on the surface area defined by diameter D.

The length "A" at which the piston moves with acceleration and the clearance between the piston 7 and the guide 9 are chosen to ensure a guaranteed cut of the diaphragm 8 along the entire perimeter, the required delay in the opening of the line passage after the diaphragm 8 is cut, the acceleration of the piston 7 necessary for the actuation of the spring retainer 13. The dimensions of the diaphragm lintels 8 are determined by the outcome of the specified pressure that ensures the destruction of the bridge.

Further, the moving shank 10 is fixed along the flow by means of a spring retainer 13, wherein the hydraulic characteristics of the open membrane assembly 4 are reproduced with high accuracy, since there are no structural elements with an indeterminate position in the flow of the component.

After opening the membrane assembly 4 due to the increased pressure of the starting fuel, the membrane assembly 5 opens in the same way.

Modifications

Differences between RD-170 and RD-171

The RD-170 and RD-171 engines use different options for swinging the chambers and their deflection controls.

The chambers of the RD-170 engine as part of Block A of the Energia rocket swing in two planes: in the radial plane passing through the longitudinal axis of the engine and the axis of the chamber, and in the tangential plane perpendicular to it. Such a control scheme is more effective in the structure of the Energia rocket package, but requires more powerful steering gears that overcome the load created by the oncoming aerodynamic flow on the protruding part of the combustion chamber nozzle beyond the parameter of the outer contour of the block when it is deflected in the radial direction.

The combustion chambers of the RD-171 engine of the first stage of the Zenit are deflected when controlled only in the tangential rolling plane. The nozzles of the chambers do not go into the streamlined stage of the aerodynamic flow and do not experience its load. Steering gears are significantly less powerful. The control efficiency of this option is sufficient for the Zenit missile.



Fig.17. RD-170
Each chamber is rocked by two steering gears (green in the photo)



Fig.18. RD-171M
Each camera is rocked by one steering gear (in the photo - to the left of the near camera)

The rest of the engine systems are unified.

RD-173 and RD-171M

At the final stage of engine development, V.P. Glushko initiated the development of a more advanced engine design, which, in comparison with the RD-170 (RD-171) engine, provided higher thrust (boosting by 5%) and in which measures should be taken to reduce the dynamic intensity of the feed units. The appropriate design documentation was developed and the engine was eventually named RD-173.

Until 1996, 28 engines were manufactured, which underwent diverse testing. The RD-173 engines use a more advanced design of feed units, primarily the main VTA. The RD-170 engine control system was seriously altered. In the process of testing the RD-173, it was confirmed that the engine start, its operation in all the envisaged modes is characterized by stable operation of all units and systems with the provision of the required starting pattern and accuracy of maintaining parameters without using oxidizer chokes. The exclusion of oxidizer chokes and, accordingly, two drives from the engine simplified its design, increased reliability and reduced the weight of the engine. The design of the bellows of the oscillating unit made of nickel alloy was introduced, which also increased the reliability of the engine.

The accumulated experience in setting up the engine control system in the process of control and technological tests using external feedback made it possible to switch to a much simpler control system consisting of two digital drives that control directly the thrust regulator and the throttle of the SDS in the process of testing the RD-173 engine. The simplification of the control system increased the reliability of the engine, reduced its weight.

In the RD-173 engine, taking into account the large positive statistics of the gas generators, the mixing heads are welded, in contrast to the flange connection in the RD-170 (RD-171) engines, where the possibility of prompt replacement of the head after the control and technological test was provided. This, as well as other solutions obtained during the development of the RD-173 engine, were used in the development of the RD-180 engine.

Orders for the manufacture of RD-171 engines ceased in 1995, but at the same time, NPO Energomash continued to manufacture a more advanced modification of the RD-170 (RD-171) engines - the RD-173 engine. Since 1995, NPO Energomash has been supplying RD-171 engines for the Sea Launch program, which were modified from the RD-170 engines previously manufactured for the first stages of the Energia launch vehicle. These engines laid the groundwork for the implementation of the program until 2004. Taking into account the accumulated experience in testing the RD-173 and RD-180 engines, in which solutions aimed at improving reliability and ensuring boosting by 5% have been introduced, NPO Energomash proposed to manufacture RD-173 engines for the Sea Launch program. This proposal was supported by Yuzhnoye, Yuzhnoye, the prime developer of the Zenit LV, and approved by the LV customer. The engine was designated RD-171M. Certification of the RD-171M engine was completed on July 5, 2004. 8 tests were carried out on the certification engine with a duration of 1093.6 seconds, and the last test (in excess of the plan) was carried out at the 105% mode. The first commercial engine RD-171M was delivered to Ukraine on March 25, 2004 after a 140-second CTI.

In 2006, the RD-171M engine was certified for use as part of the Zenit-M LV in the implementation of state programs of the Russian Federation.

Technical Diagnostics System

The system of technical diagnostics was developed in parallel with the creation of the engine as a means of assessing the technical condition engine and the forecast of its performance. In addition, it was used for failure and defect analysis because it provided an opportunity to study the interconnection of parameters and their statistical characteristics in more depth.

The system is a set of technical means, diagnostic methods and the object of diagnostics, as well as organizational and technical measures for the collection, transformation, storage, analysis of information and decision-making about the condition of the engine. The system must ensure the location and causes of malfunctions.

The technical diagnostics system has the following subsystems:

- information and measurement;
- functional diagnostics;
- test diagnostics as a non-destructive method of condition control.

In the course of the development of the diagnostic system, the following were created:

- a method for monitoring the stability of launch characteristics, main mode and final stage mode. The method was intended to evaluate the values of slowly changing parameters and their velocities obtained during firing tests, taking into account the field of permissible boundaries;
- Procedure for Tolerance Control of Parameters in the Main Mode and Terminal Stage Mode; It was intended for conformity assessment engine parameters measured during firing tests, calculated values obtained using mathematical models and model characteristics of units according to their autonomous tests, which is determined by the presence of parameters in the tolerance field;
- the method of contour linkage of slowly changing parameters; was intended to assess the functioning of the engine as a whole and its contours in stationary modes by comparing the measured and calculated values of slowly changing parameters at characteristic points;
- a method for assessing stability and determining vibroacoustic characteristics; was designed to control the level of pulsation and vibrations for compliance with statistical tolerances and assessment of the stability of the combustion chamber and gas generator, with an analysis of the physical nature of the spectra and determination of vibration attenuation decrements;
- methodology for estimating the value of the exhausted resource of assembly units; It is based on the theory of multi-cycle fatigue of materials and takes into account dynamic loads caused by pulsations and vibrations; The integral value of fatigue damage in control and technological tests, its value during operation was predicted and their sum was compared with the limit value, determined by the results of multi-resource tests;
- parametric control technique - used in diagnostics in stationary modes in order to localize malfunctions; the analysis is based on assessments of the functional characteristics of the units;
- a set of non-destructive testing methods.

In serial production, each engine, after manufacture and a full cycle of control, undergoes autonomous control technological tests, which are carried out on the manufacturer's firing stand with the engine starting according to the full flight program or a slightly accelerated one. After firing bench tests, the engine can undergo a bulkhead. This means that in order to make sure that the quality of the structure is maintained after fire tests, partial disassembly of individual units is carried out.

[Gallery](#)

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